

Slope and Rate of Change — Diagnosing Inverted and Sign-Reversed Calculations
Grade 8 / Pre-Algebra

Slope is **rise over run**, and both differences must be taken in the **same order**:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- Flipping the fraction gives the **inverted** slope (run over rise).
- Starting with point 1 on top and point 2 on the bottom gives the **sign-reversed** slope.
- In each problem, decide which of those two errors was made (if any), then compute the correct value. Show the subtraction you used.

1. Warm up. Find the slope of the line through $(2, 3)$ and $(6, 11)$. Label your rise and your run before you divide.

Answer: _____

2. Find and fix. Devi finds the slope through $(-3, 5)$ and $(2, -5)$. She writes:

$$m = \frac{5 - (-5)}{2 - (-3)} = \frac{10}{5} = 2$$

Devi's arithmetic is correct, but her answer is not. Name the error in her set-up, then give the correct slope.

Answer: _____

- 3. Find and fix.** Mika finds the slope through $(1, 4)$ and $(5, 6)$. He writes:

$$m = \frac{5 - 1}{6 - 4} = \frac{4}{2} = 2$$

Name Mika's error, then give the correct slope as a fraction.

Answer: _____

- 4.** A tank is draining. The table shows the water level, in litres, at four times.

Hours	0	2	4	6
Litres	46	40	34	28

Find the **change** in the water level from hour 0 to hour 6. A change that goes down is negative, so write your answer with its sign.

Answer: _____

5. Use the same draining tank.

Hours	0	2	4	6
Litres	46	40	34	28

Find the rate of change of the water level, in litres per hour, between hour 0 and hour 6. Keep the level difference on top and the time difference underneath, both taken in the same order.

Answer: _____

6. **Find and fix.** Ty finds the slope through (5, 9) and (1, 1). He writes:

$$m = \frac{1 - 9}{5 - 1} = \frac{-8}{4} = -2$$

Both of Ty's differences came out of the same two points, so his arithmetic looks fine — but he mixed the order. Name the error and give the correct slope. Then say whether the line rises or falls from left to right.

Answer: _____

7. A candle burns at a steady rate. At 3 minutes it is 18 cm tall, and at 11 minutes it is 6 cm tall.

Jo reports the rate of change as $\frac{11 - 3}{6 - 18} \approx -0.67$ cm per minute. Explain what Jo inverted, then find the correct rate of change of the candle's height, in centimetres per minute.

Answer: _____

8. Challenge. A line passes through $(a, 4)$ and $(7, -2)$, and its slope is -3 . Priya sets up

$$\frac{7 - a}{-2 - 4} = -3$$

and reports $a = -11$.

Diagnose Priya's set-up error, write the correct equation, and solve it for a .

Answer: _____